

AURA HOME SYSTEMS - CUSTOMER RETENTION STRATEGY

PROJECT REQUIREMENTS

Capstone Objective: Translate major elements learned pertaining to the degree program into a real-world actionable assessment. Implement an app that applies to your project that also has real-world implications.

Required Tools:

- Excel
- SQL
- Google Colab/Python
- Tableau or Power BI
- Github
- Presentation Software
- Vibe-Coding Application Creation

Note: Specifics on HOW these tools are used are completely subjective to the context of your project.

PROJECT DETAILS

Dataset: Aura Home Systems customer dataset

Goal: Customer retention strategies based on customer segmentation

Techniques:

- Data cleaning in Excel
- Machine Learning (Random Forests and K Means Clustering) in Google Colab/Python
- Querying data and additional segmenting in SQL
- Visualizing data in Power BI
- Application creation using Windsurf
- Dedicated Github repository for the project

BACKGROUND

Aura Home Systems is a fictional “Smart Home” connectivity platform that handles an array of services including phone service, internet service, online security, and online television/movie streaming. Currently Aura has lost 26.54% of their customer base valued at \$139K. They currently generate \$456K in monthly recurring charges, have an average churn probability of 27.53%, and have an at-risk loss potential of \$100K in monthly recurring charges. We are going to see how we can be proactive with retaining customers as it is 5-25x more expensive to acquire new customers than to retain one. Other stats pertaining to this matter include:

- A 5% better retention rate can generate up to 25-95% more profit.
- An existing customer has a 60-70% chance of buying again compared to 5-20% for a new prospect.
- Companies with a retention rate above 85% grow 1.8x faster than the market average.
- ROI on retention is on average 5x higher than on acquisition marketing measured over a 24-month period.
- The average churn rate for a telecom company is 22.8%. Aura is operating at 26.54%.
- The average gross revenue retention rate for a telecom company is between 80-88%. Aura currently has a GRR of 69.50%.

METADATA

(Dataset Link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn/data>)

The dataset contains customer data that can be used to predict churning, but the goal is not to use it solely for churning purposes. Our goal is to use churning probability to help us segment customers for customer retention strategies.

The dataset contains 7,043 rows and 21 columns that consist of the following columns:

- CustomerID: unique record number for each customer
- Gender: the gender of the customer
- SeniorCitizen: lets us know if the customer is a senior citizen or not
- Partner: whether the customer has a partner or not
- Dependents: whether the customer has dependents or not
- Tenure: the number of months the customer has been with the company
- PhoneService: whether the customer has phone service with the company or not

- MultipleLines: whether the customer has multiple phone lines with the company or not
- InternetService: the type of internet service the customer has with the company (if any)
- OnlineSecurity: whether the customer has online security with the company or not
- OnlineBackup: whether the customer has online backup or not
- DeviceProtection: whether the customer has device protection or not
- TechSupport: whether the customer has tech support or not
- StreamingTV: whether the customer has streaming TV or not
- StreamingMovies: whether the customer has streaming movies or not
- Contract: the contract term of the customer
- PaperlessBilling: whether the customer has paperless billing or not
- PaymentMethod: the customer payment methodology
- MonthlyCharges: the customer monthly bill amount
- TotalCharges: the total amount the customer has been billed
- Churn: whether the customer has churned or not

WORKFLOW LOGIC FOR THIS PROJECT

- Data cleaning in Excel to get a clean dataset base to work from.
- Send cleaned dataset into a Google Colab notebook to accomplish the following:
 - Churn probability for each customer (Supervised Classification – Random Forest).
 - At-risk revenue for each customer based on each customer's monthly charges multiplied by their respective churn probability.
 - Customer segmentation based on monthly charges to establish a value tier structure (K Means Clustering).
 - Exporting segmented customer data to SQL to query data for further insights.
- Send segmented customer data to SQL for further segmentation. We want to establish a tier structure based on the customers' value *and* churn probability.
- Visualize and analyze all findings in Power BI.
- After assessing all data, develop a strategy that will focus on customer retention while allocating the proper number of resources to each tier based on their combined value and churn probability.
- Finalize a report with all the data.
- Create a presentation based on findings.
- Create an app based on this dataset that will be useful in a real-world scenario.

- Send entire project to Github.

DATA CLEANING

Excel was used to clean the data and get some foundational statistical information. The findings regarding our Excel assessment are as follows:

- Process
 - Reviewed all columns to evaluate possible redundancies and/or unnecessary data and came to the conclusion that all columns were necessary at this point of the project.
 - Standardized naming conventions to the header names as all lowercase with underscores separating words.
 - Used the “Remove Duplicates” feature to remove any potential duplicates based on the customer_id column (there was no duplicates found).
 - Converted monthly_charges and total_charges to currency data types during the Excel evaluation for assessment clarity when looking at the data in its entirety.
 - Implemented COUNTA, COUNTBLANK, and COUNTA(UNIQUE) and there was no missing data in any of the columns nor was there anything out of the ordinary that stood out.
 - Implemented MIN, MAX, MEDIAN, AVERAGE as well as obtained totals for each numeric column and was given the following results:
 - “senior_citizen”
 - Should only be 0 and 1 values in this column. The MIN was 0 and the MAX was 1 which tells us everything looks as it should for this feature, eliminating outlier probability.
 - AVERAGE: 0.16
 - “tenure”
 - TOTAL: 227,990 months
 - MIN: 0 months
 - MAX: 72 months
 - MEDIAN: 29 months
 - AVERAGE: 32 months
 - “monthly_charges”
 - TOTAL: \$456,116.60
 - MIN: \$18.25

- MAX: \$118.75
 - MEDIAN: \$70.35
 - AVERAGE: \$64.76
- “Total_charges”
 - TOTAL: \$16,056,168.70
 - MIN: \$18.80
 - MAX: \$8,684.80
 - MEDIAN: \$1,397.48
 - AVERAGE: \$2,283.30
- Converted “Female” to “1” and “Male” to “0” in the “gender” column to prepare the data for SQL and Python.
- Converted “Yes” to “1” and “No” to “0” for partner / dependents / phone_service / paperless_billing / churn columns to prepare data for SQL and Python.
- After converting the “gender / partner / dependents / phone_service / paperless_billing / churn” columns to binary numeric data, I re-evaluated the MIN AND MAX of these columns to ensure that these columns only had “0” and “1,” and all of these columns only had a “0” or “1” value.
- Cleaning Conclusion
 - The MIN, MAX, MEDIAN, and AVERAGE for “tenure” and “monthly_charges” look acceptable given the context of the dataset with no major outliers to assess.
 - At first glance, looking at the MIN, MAX, MEDIAN, and AVERAGE of total_charges might seem as if there are some outliers to assess, but upon further investigation (multiplying “tenure” by “monthly_charges”), these numbers seem acceptable given the context of the dataset.
 - All in all, no major outliers to report.
 - All feature headers have been converted to a standardized naming convention.
 - No duplicate records were found.
 - All essential data is present.
 - Exported the finalized cleaned dataset as “Aura_Final” for further assessment.

MACHINE LEARNING

We imported a cleaned dataset (Aura_Final) into our Google Colab notebook to get predictive-based data that will help us with our customer retention strategy. Even though the dataset was cleaned in Excel, we wanted to make sure the data would not be skewed in any way that can affect the models we were about to implement, so we did some light cleaning and assessing as a precautionary measure.

- Light Cleaning and Structural Investigation
 - Stripped away any blank spaces where applicable to ensure all data is being read correctly.
 - Data frame shape was accurate.
 - “monthly_charges” and “total_charges” were incorrectly categorized as object data types. Converted these features into float data types.
 - Basic statistical data seemed on par with our Excel assessment.
 - Everything else pertaining to the structure and integrity of the dataset seemed accurate.
- Supervised Classification – Random Forest
 - We are using a Random Forest model to predict churn probability for each customer in our dataset because of the Random Forest model’s ability to handle complex datasets without overfitting.
 - Random Forest Process:
 - Data split between training and testing datasets.
 - “Dummy” creation for categorical columns that have not been converted to numerical data.
 - Scaled numerical features.
 - Implemented hyperparameter tuning to get the best possible results.
 - Executed the model and received the following results according to the classification report and confusion matrix:
 - Classification Report
 - Class 0 (non-churners):
 - Precision: 0.91
 - Recall: 0.76
 - F1-Score: 0.83
 - Class 1 (churners):
 - Precision: 0.54
 - Recall: 0.78
 - F1-Score: 0.64
 - Classification Report Conclusion:

- Class 0 (Non Churners) - For the most part the model is very good at recognizing customers who will stay with a precision of 0.91 and a recall of 0.76. An F1 score of 0.83 indicates a strong overall performance for Class 0.
 - Class 1 (Churners) - The model is solid when it comes to catching churners with a recall of 0.78 but lacks a bit with a precision of 0.54. Seems to be the trade off of having a solid recall of 0.78 in a churning scenario. F1 score is a solid balance at 0.64.
 - The model predicted 78% of all customers who actually churned (Class 1 recall) which could be considered the “trade-off” for the low precision score of 54% when keeping in mind that catching churners is more important than avoiding false alarms.
- Confusion Matrix:
 - True Negatives: 791
 - True Positives: 292
 - False Negatives: 82
 - False Positives: 244
 - Confusion Matrix Conclusion:
 - True Negatives: The model correctly identified 791 customers who stayed.
 - True Positive: The model correctly identified 292 customers who churned.
 - False Negatives: The model incorrectly identified 82 churners as non-churners.
 - False Positives: The model incorrectly identified 244 non-churners as churners.
- Random Forest Conclusion:
 - The recall vs precision trade-off is acceptable in this scenario as we are looking to catch churners rather than miss them. It is a solid model all-around. We will now further assess the model by examining the weight of each feature importance.
- Feature Importance Weight Validation:

- We wanted to evaluate the weight of each feature used in the model as a precautionary measure. After scoring each feature, the top 10 features with their respective weight value are as follows:
 - 1. tenure:
 - Weight: 0.1753 (17.53%)
 - 2. total_charges
 - Weight: 0.1322 (13.22%)
 - 3. contract_Two year
 - Weight: 0.1244 (12.44%)
 - 4. monthly_charges
 - Weight: 0.1135 (11.35%)
 - 5. internet_service_Fiber optic
 - Weight: 0.0969 (9.69%)
 - 6. payment_method_Electronic check
 - Weight: 0.0597 (5.97%)
 - 7. contract_One year
 - Weight: 0.0473 (4.73%)
 - 8. internet_service_No
 - Weight: 0.0468 (4.68%)
 - 9. online_security
 - Weight: 0.0282 (2.82%)
 - 10. paperless_billing
 - Weight: 0.0259 (2.59%)
 - Feature Importance Weight Conclusion:
 - Given the context of project and upon further examination of the features it looks as if the weight distribution is fairly reasonable for each feature. We will now proceed with implementing at-risk predictions for each customer in the dataset.
- At-Risk Predictions:
 - After obtaining an acceptable model and evaluating its respective features, we moved on to getting the at-risk prediction metrics we need for our customer retention strategy. After scaling the numeric features we were able to process the data in a way that gave us the “churn_probability” of each customer. We were then able to use the “churn_probability” to give us the “at_risk_revenue” for each customer.

- Segmenting Customers with K Means Clustering:
 - We need to establish tiers for targeted marketing campaigns for at-risk clientele. To accomplish this, we need to segment the customers so that we can develop a customer retention strategy for each respective customer group.
 - Once the customers were segmented into their respective clusters, we were able to place them as follows:
 - Tier 1 (Low): These are the customers who spend the least on monthly charges. There are 1,892 customers who belong to this tier. They have a minimum monthly charge value of \$18.25, a maximum of \$42.40, with an average of \$23.38.
 - Tier 2 (Medium): These are the customers who spend the medium amount of money on monthly charges out of the entire dataset. There are 2,239 customers who belong to this tier. They have a minimum of monthly charge value of \$42.60, a maximum of \$77.80 with an average of \$61.63.
 - Tier 3 (High): These are the customers who spend the most on monthly charges. There are 2,912 customers who belong to this tier. They have a minimum of monthly charge value of \$77.85, a maximum of \$118.75, with an average of \$94.05.
- We will now export the new version of this dataset into SQL for further investigation and segmentation. The name of this new version of this dataset will be “Aura_Churn_Tiers.”

SQL

We used SQL to query and segment data that will be useful for our overall assessment. We created a database called “Aura_Analytics” and uploaded our “Aura_Churn_Tiers” table inside of the database.

- Upon uploading the dataset as a table into the database we used the following process:
 - Right clicked on the database → Tasks → Import Flat File
 - Used the import wizard to upload the data into SQL.
 - Changed “customer_id” to varchar50 data type in both tables.
 - Allowed null values in both tables to ensure the uploading of both datasets into the database.

- Querying summarized churning-based data from the original (finalized) dataset/Aura_Cust_Metrics table:
 - Total Customers: 7,043
 - Sum of the customer_id column
 - Total Customers Lost: 1,869
 - Sum of the customers with a “1” in the “churn” column
 - Churn Rate Percentage: 26.54%
 - $\text{Total Customers Lost} / \text{Total customers} \times 100$
 - Lost Monthly Recurring Revenue: 139,130.85
 - Sum of “monthly_charges” based on every customer with a “1” in the “churn” column.
 - These results were exported from SQL into a spreadsheet file named “Aura_Original_Totals” as a reference table.
- Next we created a new column called “churn_tiers” where we categorized each customer has either “High Risk,” “Med Risk,” or “Not Targeted.” The ideology for this tier structure is as follows:
 - Customer with a 75-90% risk of churning are considered “High Risk.”
 - Customers with a 30-75% risk of churning are considered “Med Risk.”
 - Customers below a 30% risk of churning and customers above a 90% risk of churning are considered “Not Targeted” customers. This is because it is highly unlikely that a customer below a 30% risk of churning will actually churn and customers above a 90% risk of churning are very likely to churn and are considered “lost causes.” Given that we have to allocate our resources accordingly, we will not be targeting customers below a 30% risk of churning or above a 90% risk of churning.
- Essentially we are only accounting for current customers with a 30-90% chance of churning and further segmenting them for our customer retention strategy.
- I exported the updated dataset with the newly created “churn_tier” column into a new dataset called “Aura_Updated_Final.”
- After assessing the segmented data, I decided to consolidate the segments into three distinct tiers to better focus our efforts:
 - Tier 1: Low Value/High Risk + Low Value/Med Risk
 - Tier 2: High Value/Med Risk + Med Value/Med Risk
 - Tier 3: High Value/High Risk + Med Value/High Risk

POWER BI

Four dashboards were created in Power BI based on the Aura_Updated_Final dataset. We also created custom measures to properly represent our findings within the dashboards.

- We created a new table in Power BI named “New Fields” to account for the measures that were created. The measures are as follows:
 - ARPU (Average Revenue Per User): $\text{Sum of total monthly charges} / \text{total number of customers} = \text{ARPU}$
 - Churn Rate %: $\text{Sum of total number of churners} / \text{total number of customers} = \text{Churn Rate \%}$
 - Avg CLV (Average Customer Lifetime Value): $\text{Average of total monthly charges of non-churners} / \text{Churn Rate \% of non-churners} = \text{Avg CLV}$
 - Max Recoverable Revenue: Sum of monthly charges for all non-churners that have a 30-90% churn probability
 - Segment Campaign Cost: $(\text{Sum of monthly charges for non-churners} * 7\%) * 40\% = \text{Segment Campaign Cost}$
 - Considered a standard customer retention formula for a telecom company’s campaign costs in its “growth” stage.
 - Expected Recoverable Revenue: $(\text{Max Recoverable Revenue} * 25\%) - \text{Segment Campaign Cost} = \text{Expected Recoverable Revenue}$
 - Considered a standard customer retention formula for a telecom company’s campaign costs in its “growth” stage.
 - GRR (Gross Retention Revenue Rate): $\text{Sum of monthly charges for non-churners} / \text{sum of monthly charges for all customers} = \text{GRR}$
 - Retention ROI: $\text{Expected Recoverable Revenue} / \text{Segment Campaign Cost} = \text{Retention ROI}$
- Dashboards:
 - Numbers at a Glance: This dashboard was designed to give the viewer an overall state of the business regarding customer value and loss due to churning. This dashboard contains the following:
 - Churn Rate % card: 26.54%
 - Total MRR card: \$456,116.60
 - Avg CLV card: \$169.60
 - Churn Probability card: 27.41%
 - Total Number of Customers card: 7,043
 - ARPU card: \$64.76
 - Churned Customers vs Current Customers donut chart
 - Lost MRR vs Maintained MRR donut chart
 - Churn Probability histogram

- Retention Risk and Reward: This dashboard gives the viewer an overall idea of what is risk vs reward implications of customer retention. This dashboard contains the following:
 - Retention ROI: 792.86%
 - Max Recoverable Revenue: \$143,461
 - Expected Recoverable Revenue: \$31,848.26
 - Total At-Risk Revenue: \$100,100.25
 - GRR: 69.50%
 - Campaign Cost: \$4,016.89
 - Retention Opportunity Snapshot column chart with Expected Recoverable Revenue vs Max Recoverable Revenue
 - At-Risk Revenue vs Tenure by Contract Type scatter plot
- “Tiers” by the Numbers: This dashboard gives the viewer an overall assessment of the tier by the numbers as well as a breakdown of each individual tier. This dashboard contains the following:
 - Total Tiers Max Potential MRR Loss card: \$143,461
 - Total Tiers At-Risk Revenue card: \$77,257
 - Total Tiers Avg Churn Probability: 53.5%
 - Total Tiers Avg Customer Lifetime Value: \$351.62
 - Total Tiers Avg Revenue Per User: \$73.42
 - Tier 1, Tier 2, and Tier 3 Churn Probability gauge charts (one gauge per tier)
 - At-Risk Revenue vs Max Potential MRR Loss by Tier horizontal bar chart
 - ARPU vs CLV By Tier horizontal bar chart
 - Min vs Max Monthly Charge by Tier horizontal bar chart
 - Customer Count by Tier donut chart
- Current Customer Reference Table: This dashboard is a reference point for granular data if the viewer wants specifics. This dashboard contains the following:
 - A Text Table with the following:
 - Customer ID: This uniquely identifies each customer.
 - Tier: The tier that the customer belongs to based on their monthly charge amount and churn probability combination.
 - Monthly Charges: The actual amount of monthly charges each customer pays.
 - Churn Probability: The churn probability of each customer.

- At-Risk Revenue: The amount of revenue that is at risk for each customer based on their churn probability and monthly charges (churn probability * monthly charges = at-risk revenue).
- Contract Type: The contract type for each customer.
- We have three slicers that are associated with this visual to assist the viewer in finding the data they are looking for. The slicers are as follows:
 - Tiers
 - Contract Type
 - Churn Probability

CUSTOMER RETENTION STRATEGY

After assessing the segmented groups, we came up with a retention strategy for each group based that more or less revolves around a universal retention strategy.

- Universal Retention Strategy
 - Referral program with a \$25 bill credit for each successful referral
 - \$5 monthly bill credit for enrolling in automated payments
 - Personalized “Thank You” message
 - Customer out-reach to evaluate the quality of their experience on a periodic basis
 - Free equipment cleaning and diagnostic check
 - Lots of customers churn due to poor performance issues. Ensuring that the equipment is working properly is a very effective measure in deterring customers from churning.
 - Loyalty, point-based rewards program
 - 67% of program members spend more vs non-members
 - 28% higher purchasing frequency among members
 - 79% of customers say loyalty programs influence their decisions
 - 77% of customers say a good loyalty program makes them loyal to a brand even if a competitor is cheaper
 - The overall success rate of a loyalty program is 84%
- Tier 1 - Low Value / High Risk + Low Value / Med Risk: Low value so incentives must be low-cost and scalable.
 - \$5 off one-time bill credit
 - Free month of a streaming service add-on

- Prepaid 6-month bundle options
 - Free modem upgrade
 - Free Wi-fi extender
- Tier 2 - High Value / Med Risk + Med Value / Med Risk: Not in immediate danger but are high value, nonetheless. They will receive all Tier 1 benefits and automatic enrollment into the loyalty rewards program where they can attain points based on the following:
 - On-time payments
 - Referrals
 - Autopay enrollment
 - Downloading the app
 - Completing a customer profile
 - Participating in surveys
 - Note: Points can be redeemed for bill credits and/or upgrades.
- Tier 3 - High Value / High Risk + Med Value / High Risk: High churn probability and meaningful impact. This group needs immediate personalized human intervention! This group receives all Tier 1 and 2 incentives plus the following:
 - Priority customer support
 - Faster response service
 - Early access to new plans
 - Free month of service after 12 months
 - Waived activation/upgrade fees
 - Free add-on streaming service for 3-6 months
 - Data cap increase
- Additional Strategy – Within these tiers, newer month-to-month customers are the highest “at-risk.” We could offer the following to retain them into a more solidified contract structure:
 - Plan optimization suggestions
 - Cheaper plan alternatives with an extended commitment
 - Add-on bundles
 - 5% off annual plan

SUMMARY

Out of the \$456K monthly recurring revenue, Aura Home Systems has lost \$139K to churning. With the current 26.54% churn rate percentage and 27.41% churn probability percentage, Aura Home Systems is at a risk of losing over \$100K in revenue with their

current customer base. By implementing a customer retention strategy where we are segmenting “attainable” and “at-risk” clientele, we can expect to save at least \$32K with a \$4K investment, giving us a retention ROI of 792.86%.

FURTHER RESEARCH/NEXT STEPS

- Date specifics for the dataset. Having an idea of the time of year when customers are most likely to churn can help us understand seasonal churning and can help us with our retention strategy.
- Name, address, and email information for each customer included in the dataset. This can streamline multi-department usage.
- Customer address information can also help us see if there are any regional patterns that we need to watch out for.
- Preferred method of contact for each customer to further understand customer behavior when assessing at-risk potential.
- Discuss specifics of the point-based rewards loyalty program.
- Interdepartmental meeting with accounting, marketing, and staff managers to further discuss the cost particulars of the retention strategy and how to allocate the campaign cost budget effectively.

APP CREATION

We created a Streamlit app using Windsurf based on the customers who have not churned yet, but have a 30-90% churn probability. The idea is to follow up on our retention efforts. The app is designed to enter a “Yes” or “No” in the “Retained” section to notify us if the customer was able to be retained. The app then counts the number of retained and churned customers and converts these numbers into percentages. This tells us the success rate of our efforts. This is important because if our retention efforts are not working, we need to pivot and try other retention strategies. The app also allows us to filter by each field from our Aura_Tiers dataset so that we can narrow the scope of retention attempts where applicable.

REFERENCES

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